Manual

Standard Control Charts and Histograms

WEP-RKH 8.1

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Copyright

Contents

1 Standard Control Charts and Histograms - Overview

Purpose

This component expands the functions of the inspection requirements to

- display control charts and histograms.

Implementation Considerations

If the inspection requirements functions require order related graphical display of the values over time, use of this component is recommended.

Integration

This component mainly serves the component "Goods Receipt Inspection Planning".

Features

Displaying a control chart and histogram for each inspection order characteristic to clearly visualize the collected inspection data (e.g. measured values).

2 Inspection Requirements

Summary

Menu	Quality management → In-production inspection → Inspection requirement Quality management → Goods receipt → Inspection requirement Quality management → Goods issue → Inspection requirement Quality management → Initial sample → Inspection requirement Quality management → Gage management → Inspection requirement
Transaction code	irp[1]
Function authorization	irp

Utilization

The inspection requirement is as important as the production order is in the shop floor data collection module and includes one or several inspection steps (e.g. one inspection step for each operation). In this context, inspection steps result from the inspection plan search triggered by the generation of inspection requirements. For this reason, the inspection requirement rather includes general information, such as the order or article number. The inspection step that is subordinate to the inspection requirement finally includes the characteristics to be checked. Consequently, the inspection step or the operation of the production order (and, as a result, the inspection step as well) is selected and logged on to the terminal for measurement recording.

As the functional requirements are nearly identical in all areas (e.g. goods receipt, production, goods issue, gage management), the corresponding applications are structured nearly in the same way for these areas.

[1]New inspection requirements cannot be created if the "inspection requirement" application is opened by using transaction codes.

In creating inspection requirements, the user is generally supported by master data catalogs. For this reason, it is very important to conscientiously maintain master data.

Integration

Inspection requests/inspection steps are generated on the basis of inspection plans that have been created beforehand and refer to them. When being generated, the contents of inspection plans are copied into the inspection request/inspection steps. Thus, the relation is only based on referencing. The limited modification options that are enabled in the inspection plan do not affect already generated inspection requirements/inspection steps.

The inspection steps, in turn, are the "objects" that can be checked/inspected and the operation number connects them with the corresponding operation of the production order within production. Within gage management, inspection steps can be put on a level with the actual calibration order.

Nearly all reports/evaluations refer to the data of inspection requests/inspection steps and the available inspection step characteristics. Restricting filters are, e.g. the article number, order number, operation, and inspection step characteristic.

In gage management, inspection requirements are generated automatically from the activity calendar by a user action. In exceptional cases, inspection requirements can also be generated directly in the “inspection requirement” application. In this case however, the inspection plan number has to be entered manually.

Prerequisite

Inspection plans need to be created beforehand in the same area/context to be able to create inspection requests/inspection steps. Master data is also required. Which master data has to be maintained depends on the respective field of application.

Selection criteria

The list that follows shows some of the available selection criteria. Self-explanatory filter options are not listed.

Area

Selection list of the configured areas of in-production inspection, goods receipt or goods issue. By default, the following areas are available:
In-production inspection: Production
Goods receipt: Goods receipt
Goods issue: Goods issue
Initial sample: Initial sample inspection
Gage management: Gage management

Inspection request number

The inspection requirement number uniquely identifies the inspection request. HYDRA automatically generates the inspection requirement number, which is unique.

Status

By opening a status list, the different inspection request statuses may be filtered. In this way, it is possible to restrict the list of inspection requirements, e.g. to the completed inspection requests.

Skip lot (available for the goods receipt and goods issue only)

Due to dynamic modifications, an inspection requirement can get the classification "skip lot" and, as result, it will not be checked.

Order

Number of the corresponding production order or calibration order.

Article number

Direct input or opening of the article catalog and taking over of the number from the entry selected there.

Drawing issue number

The article is identified uniquely by the combination of "article number" and "drawing issue number".

Customer number

Direct input or opening of the customer catalog and taking over of the number from the entry selected there.

Supplier number

Direct input or opening of the supplier catalog and taking over of the number from the entry selected there.

Manufacturer number

Direct input or opening of the manufacturer catalog and taking over of the number from the entry selected there.

Actual date from / until

The time may be restricted here. Normally, in production this field is not filled out. In the goods receipt area the "actual date" is the date when the goods are delivered.

Planned date from / until

The time may be restricted here. Normally, in production this field is not filled out. In the goods receipt sector the "planned date" defines the date when the goods are (initially) planned to be delivered.

Manufacturing date from / until

The time may be restricted here.

"Inspection request" field descriptions

The elementary fields are described below.

Inspection requirement

Area

Area in which the inspection request is to be created. A selection list of the configured areas for in-production inspection, goods receipt, goods issue, initial sampling or gage management facilitates the creation process. By default, the following areas are available.

In-production inspection: Production
Goods receipt: Goods receipt
Goods issue: Goods issue
Initial sample: Initial sample inspection
Gage management: Gage management

This field is a key field for searching for the active inspection plan or the specified inspection plan including index on the basis of which the inspection request, inspection steps and inspection step characteristics are to be generated.

Inspection request number

The inspection request number is automatically generated upon saving. It is numeric, identifies an inspection request uniquely and cannot be changed.

Operation, operation designation

If operations are in use only one of the two fields "operation" or "operation designation" has to be filled out. Only the “operation number” field is to be filled out, provided that an inspection requirement and/or an inspection order is to be generated automatically by logging an operation of a production order on. The work plan structures cause the operation number to be referenced almost exclusively.

Subject to system configuration, this field can be a key field for searching for the active inspection plan on the basis of which the inspection requirement, inspection steps and inspection step characteristics are to be generated. If the system is configured accordingly, all inspection steps relating to the operations included in the relevant/detected inspection plan are generated already when the inspection requirement is generated.

Even areas that are not assigned to a production order (e.g. goods receipt and gage management), might require a “fictitious” operation (e.g. 9999) to be specified, subject to system configuration. This “fictitious” operation has to be included in the relevant/detected inspection plan.

Inspection plan number / inspection plan index (specified)

As an alternative to indicating key fields for searching for the active inspection plan to generate inspection requirements, the inspection plan that is to be used for the generation may also be entered in these fields.

Date

When a new data record is created, this field includes the system date, which may be changed afterwards. Once it has been saved, this field may no longer be changed.

Article

Input of the article number. Provided that it is known, it may be entered directly. Otherwise, the article catalog can be opened and the requested article can be identified and taken over using the filter and sort criteria provided there. By selecting an article, the drawing issue number, article designation, customer article number and the drawing number are taken over from the master data record and displayed in the corresponding fields.

This field is a key field for searching for the active inspection plan on the basis of which the inspection requirement, inspection steps and inspection step characteristics are to be generated, provided that no inspection plan including index has been specified.

This field is empty or should be left empty in gage management.

Drawing issue number

Like the article number, the drawing issue number can be entered directly.

This field is a key field for searching for the active inspection plan on the basis of which the inspection requirement, inspection steps and inspection step characteristics are to be generated, provided that no inspection plan including index has been specified.

In case the article number and the drawing issue number are entered directly, the master data record is determined on the basis of this information while saving and the article designation, customer article number as well as the drawing number are displayed.

The article is identified uniquely by the combination of article and drawing issue number. It is not obligatory to define and use a drawing issue number.

Provided that the “drawing issue number” field is shown in gage management, it is not allowed to assign any values to it here.

Companies

Customer / supplier / manufacturer

The corresponding company number may either directly be entered or by opening the corresponding master data catalog and selecting the corresponding entry. Once the inspection requirement has been selected and/or saved, the company name and address data are displayed additionally.

These fields are key fields for searching for the active inspection plan on the basis of which the inspection request, inspection steps and inspection step characteristics are to be generated, provided that no inspection plan including index has been specified.

Date - quantities

Actual quantity

The inspection scope is calculated subject to the selected sampling scheme and the actual quantity. This is especially true in goods receipt and goods issue if dynamic modification is enabled. Normally, in production this field remains empty.

Target quantity

In particular for the goods receipt, it is important to indicate the target quantity in order to determine the difference between the ordered quantity and the actually delivered quantity.

Scrap

This field is only relevant if HYDRA exclusively uses the CAQ functions and the shop floor data collection (BDE) module is not activated.

Rework

This field is only relevant if HYDRA exclusively uses the CAQ functions and the shop floor data collection (BDE) module is not activated.

Actually produced quantity (act. prod. qty.)

This field is only relevant if HYDRA exclusively uses the CAQ functions and the shop floor data collection (BDE) module is not activated.

Actual date

In particular for the goods receipt, it is important to indicate the actual date in order to determine the difference between the planned delivery date and the actual delivery date. Normally, in production this field remains empty.

Planned date

In particular for the goods receipt, it is important to indicate the planned date in order to determine the difference between the planned delivery date and the actual delivery date. Normally, in production this field remains empty.

Status - details

Status

The status describes the inspection requirement status, e.g. whether it is completed, unchecked ("created" status) or whether inspection results already exist ("partial result" status).

Overall result

The overall result defines whether the inspection request is classified as "pass" or "fail".

Usage decision

The usage decision classifies the overall result in more detail. The list of usage decisions can be complemented when customizing the system. The usage decision is preset subject to the overall result when inspection requirements are completed. A "pass" result leads to the usage decision "release" and a "fail" result leads to the usage decision "reject". The preset usage decision may be changed when completing the inspection requirement manually.

Skip lot

Due to dynamic modifications, an inspection requirement can be classified as "skip lot" and, as a result, it will not be checked.

PPS/ERP reference

If inspection requirements are automatically generated by higher-level systems using an interface, this reference number allows for unique assignments to be made for uploading the results.

PPS/ERP addition

If inspection requirements are automatically generated by higher-level systems using an interface, another piece of information may be added to the unique reference number for uploading results. This field is only filled out by interface.

Uploaded to PPS/ERP

If this field is enabled (by interface) it indicates that the upload to the higher-level system has been performed and the higher-level system has confirmed this via interface. This field is only populated by interface.

Inspection plan number/inspection plan index (used)

The inspection plan number and inspection plan index make clear which inspection plan was used as basis for the generation of the inspection requirement and subordinate data (inspection steps, inspection step characteristics).

Cavity assignment

The authorization "ipl_cav" is required for displaying these fields.

The options "none" or "sample" are available for cavity assignment. These options derive from the inspection plan.

Dynamic modification

Dynamic modification type

Shows the dynamic modification type of the respective inspection requirement. .
Batch-related: The available inspection plan has a dynamic modification based on batches.
Characteristic-related: The existing inspection plan has a dynamic modification based on characteristics. In this case, the transitional definition, the initial inspection severity as well as the dynamic modification norm are kept on the level of inspection step characteristics.
None: The available inspection plan does not include dynamic modification.

Transitional definition

These fields (number and designation) are only available if dynamic modification based on batches or characteristics is defined in the existing inspection plan. The transitional definition provides the possible inspection severities and controls switching between these different inspection severities. The content cannot be changed.

Determined inspection severity

It defines which inspection severity is/was used for checking the goods received according to the basics for the dynamic modification history. This field is only available if dynamic modification based on batches is defined in the existing inspection plan. The content cannot be changed.

Revised inspection severity

It defines which inspection severity is/was used for checking the goods received according to revising the inspection severity manually. This field is only available if dynamic modification based on batches is defined in the existing inspection plan.

Form

Party in charge type

Provides the shortlist of the different categories of responsible parties, e.g. department, customer, supplier, external persons

Party in charge

List of the responsible parties restricted to the type selected beforehand. This list includes all the master data that were marked accordingly in the “party in charge” field.

Party in charge name

Only shows the name/designation of the selected responsible party.

Form

Form type of the available inspection plan.

Calibration

Resource type

Shows the resource type of the resource to be calibrated. For gages, this is the type “PRM”.

Resource

Shows the resource number (gage number) that is assigned to this inspection requirement.

Maintenance

Shows the activity number that is assigned automatically by creating an activity in the activity calendar. The relevant activity is determined or referenced based on this unique ID number.

Calibration findings

Shows the result entered or determined by completing the calibration process, e.g. capable without restrictions

Calibration status

Shows the status entered or determined by completing the calibration process, e.g. released

Inspection requirement - editing functions

The key fields "area" and "inspection requirement number" as well as the fields "skip lot" and "uploaded to ERP/PPS" cannot be changed in the editing mode.

Inspection requirement toolbar

Release

Function authorization: irp.release

Transfers the inspection requirement to the status that existed at the time when it was completed, e.g. to the "partial result" status. This is required in cases where the inspection request has already been completed and is to be brought again into a condition that can be checked.

Complete

Function authorization: irp.complete

Opens a dialog to assign a usage decision, whereas a usage decision is already preset on the basis of the existing inspection results. The inspection request is finally transferred to the "completed" status, once it has been completed. The inspection steps pertaining to it can no longer be checked in this status.

Cancel

Function authorization: irp.cancel

Transfers the inspection requirement to the "canceled" status. This status no longer allows for the inspection steps pertaining to it to be checked.

"Print" detail application

Function authorization	irp.print

The print dialog of the inspection requirement header opens a list of available reports. These are Word forms. The potential content of these forms is determined by the Web services that are available in the respective context. The form entries, i.e. the content of the list of forms of the corresponding print dialog, are defined within the master data of quality management. This is also where the basis for new forms is established and the corresponding form properties are defined. A corresponding license is required to be able to change the forms with respect to content and design.

The report, 4th edition, 2nd volume VDA (German Association of the Automotive Industry), is available as special form for initial sampling. In addition to the cover sheet, the relevant annex sheets of the inspection characteristic results are printed as well. Each appendix includes the inspection characteristics including the relevant results for a category, e.g. data for dimensional checks (category: dimension).

Print - toolbar

There are no other special function buttons in addition to the standard functions/features.

"Complete inspection requirement“ detail application

Function authorization	irp.complete

The corresponding button has to be clicked in the toolbar to complete an inspection request. An editing dialog opens afterwards.

Information about the inspection steps pertaining to the inspection requirement is displayed in addition to information about the inspection requirement itself (i.e. order, batch, article, company data). An inspection requirement cannot be completed as long as any inspection steps are still open, i.e. the field “without result” includes a value greater than zero. Within this dialog box it is, however, possible to complete all inspection orders that are still open by using the button “complete all inspection orders”, provided that the conditions of a mandatory inspection that might exist are met. The result and further use of the inspection requirement may then be specified. If all of the inspection steps have been completed with “pass”, the system proposes “pass” as the result and “release” as the possible use. If any inspection step has been completed with “fail”, the system proposes "fail” as the result and “reject” as the possible use. Another result and use may still be selected though.

The list indicating the available usage decisions may be changed through MPDV customizing services.

"Inspection step" detail application

Proceeding from an inspection requirement in the master detail grid to an inspection requirement in the sub-level of "inspection steps" results in all inspection steps pertaining to the selected inspection requirement to be displayed. The detailed view of the selected inspection step additionally shows the inspection step data. The number or the combination of the inspection steps depends on the configurations made for the existing inspection plan. In case an inspection plan includes a configuration saying that the operation is defined on the level of inspection plan characteristics and, in addition to this, system configuration provides for the creation of all corresponding inspection steps along with the generation of the inspection requirement and the first inspection step, the inspection requirement includes an inspection step for every operation included in the inspection plan.

Moreover, it is possible to release, complete or cancel an inspection step using the toolbar functions.

Inspection steps may be deleted but not created or edited manually.

The current status of the inspection step can be taken from the "rating" category of the "inspection step" tab. The following statuses are possible.

Status	Description
No characteristic (KMM)	The inspection step has no characteristics able to be inspected. The inspection step is faulty and no inspection can be performed.
Created (ERS)	The inspection step has been created but not yet released for inspection. No measurement values have been recorded yet.
Inspection step released (FRE)	The inspection step has been released for inspection. No measurement values have been recorded yet.
Partial result (TER)	The inspection step has been released; some measurement values have been recorded.
Inspection in progress (IPR)	Measurement values for this inspection step are currently being recorded at another terminal.
Completed (ABG)	The inspection step has been completed.
Canceled (STO)	The inspection step has been canceled.
Skip lot (SKL)	The inspection step is in the "skip-lot" state.

"Inspection step" field descriptions

The essential inspection point fields are described in the paragraphs that follow. Self-explanatory fields are not explained.

Inspection step

Inspection step number

Unique number of the inspection step

Status

Status description of the inspection step, e.g. "completed", "partial result", etc.

Result

Qualitative classification of the inspection result based on the corresponding inspection results of the characteristics pertaining to it.

Workplace

Number of the workplace for which this inspection step has been generated.

Workplace designation

Name of the workplace for which this inspection step has been generated.

Inspection point identification

Up to nine different fields may be activated, which can or must be filled out later when inspection points are generated. By customizing the CAQ system options, it may be defined which fields are to be enabled, how they are labeled (for some fields) and whether or not it is a mandatory field.

If a field is activated and it is a mandatory field the corresponding checkbox will be "checked". The corresponding checkbox is "checked" and grayed out if it is not a mandatory field but enabled for the resulting data collection. If it is disabled, the checkbox of the field pertaining to it is not activated at all.

Once activated within the inspection point, the fields 1 to 3 are available for entering values with respect to "equipment", "functional location" and "physical sample". The fields 4 to 9 are user fields relating to inspection points. The fields 4 and 5 are "alphanumeric" fields, the fields 6 and 7 are numeric, the field 8 is a "date" field and field 9 is a "time" field.

Order data for the inspection step

Order

This is e.g. the production order number in production.

Operation

Number of the operation for which this inspection step has been generated. The operation number taken from the existing inspection plan is the basis in this context.

Operation designation

Operation designation pertaining to the operation number taken from the existing inspection plan. Normally, this field remains empty, as in nearly every case only the operation number is used in inspection planning. The operation designation is only used in exceptional cases.

Quantities

Rework

A rework quantity may be entered in this field when the inspection step is completed. This quantity specification is used for informative purposes only and is not processed by default. Normally, this field is only used if the shop floor data collection module (BDE) is not used in production, i.e. only the CAQ module is in use. Moreover, this field is useful in the goods receipt and goods issue area.

Scrap

A scrap quantity may be entered in this field when the inspection step is completed. This quantity specification is used for informative purposes only and is not processed by default. Normally, this field is only used if the shop floor data collection module (BDE) is not used in production, i.e. only the CAQ module is in use. Moreover, this field is useful in the goods receipt and goods issue area.

Actually produced quantity (act. prod. qty.)

The actually produced quantity may be entered in this field when the inspection step is completed. This quantity specification is used for informative purposes only and is not processed by default. Normally, this field is only used if the shop floor data collection module (BDE) is not used in production, i.e. only the CAQ module is in use. Moreover, this field is useful in the goods receipt and goods issue area.

Inspection step - editing functions

An inspection step cannot be edited. Only quantity specifications may be added while completing inspection steps.

Inspection step - toolbar

Release

Function authorization: irisp.release

Transfers an inspection step of the status "created" or "completed" to the "released" status or to the status that existed at the time when it was completed, e.g. to the "partial result" status.

Complete

Function authorization: irisp.complete

Opens a dialog where the (inspection) result is preset on the basis of the existing inspection results of the characteristics. By completing the inspection step, it is finally transferred to the "completed" status. Inspections can no longer be performed for inspection steps in this status.

Cancel

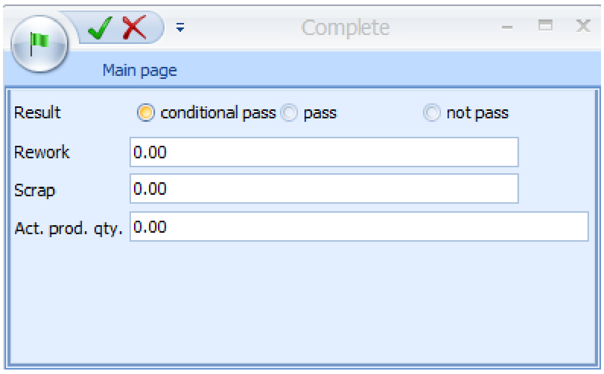
Function authorization: irisp.cancel

Transfers the inspection step to the "canceled" status. Inspections can no longer be performed for inspection steps in this status.

"Complete inspection step" detail application

Function authorization	irisp.complete

The corresponding button has to be clicked in the toolbar to complete an inspection step. The following editing dialog opens afterwards.



The system suggests the result "pass" if all corresponding inspection step characteristics have the inspection result "pass" or if they are not yet checked. The system suggests the result "fail" if there is at least one inspection step characteristic that has the inspection result "fail". However, the user is free to choose another result.

The "rework quantity", "scrap quantity" and the "actually produced quantity" may be entered when the inspection step is completed (irrelevant to gage management). This quantity specification is used for informative purposes only and is not processed by default. Normally, this field is only used if the shop floor data collection module (BDE) is not used in production, i.e. only the CAQ module is in use. Moreover, these fields are useful in the goods receipt and goods issue area. In particular in the goods receipt area, it is useful to record scrap and rework as these details may be uploaded for several inspection steps to a higher-level system at a later point in time when inspection requirements are completed.

"Inspection points" detail application

The standard configuration of the system specifies that inspection points are only used in the production area. Using inspection points only makes sense if several samples are drawn and checked within an order/inspection requirement. This is, for example, the case if inspections are to be performed subject to different events in production. This might, for example, be the case if a specified inspection interval based on quantities or time is reached. As the individual inspection plan characteristics might have different events that make inspections become due, the inspection points of an inspection step might indeed have different inspection characteristics.

Once an inspection has been performed, every inspection point is to be completed. Every inspection point has an inspection result (pass, fail) and a status (completed, open). Inspection points that have not yet been checked have a "pass" inspection result by default.

Please note

If text exceeding the field lengths existing in the relevant inspection point dialog is entered in inspection point fields that are also visible/used on AIP terminals, this field content will be cut off when saving the inspection point in AIP. Therefore, the field contents should not be longer than the field lengths configured for the AIP inspection point dialog. The field lengths of AIP dialogs can be adjusted as part of system customization.

"Inspection points" field descriptions

The essential inspection point fields are described in the paragraphs that follow. Self-explanatory fields are not explained.

Inspection point

Inspection point number

The inspection point number identifies the inspection point uniquely among the inspection points pertaining to the inspection step. The superordinate key fields "inspection step number" and "inspection request number" are shown in addition to the "inspection point" field.

Inspection result

The inspection result can be "pass" or "fail". Inspection points that have not yet been checked have the inspection result "pass" by default (standard system configuration).

Status

An inspection point can be completed or open. Only open inspection points may still be checked at the AIP terminal.

Inspection point identification

The fields, which have been identified as being activated in the superordinate inspection step, are shown here. The CAQ system options define which fields identify the inspection point and which ones require input. The following fields are displayed by default, whereas none of these fields requires input.

Date

Allows for a date to be entered. The system date is preset in case new inspection points are created. This date may directly be changed if they are created manually. Otherwise, the editing mode allows for changes to be made.

Time

Allows for a time to be entered. The system time is preset in case new inspection points are created. This time may directly be changed if they are created manually. Otherwise, the editing mode allows for changes to be made.

In addition to the configurable identification fields, there are still the following identification fields that are always available.

Workplace

If the inspection point is generated at an AIP terminal or the automatic generation is triggered by an inspection event of a specific terminal, this field shows the workplace of the triggering inspection station. If an inspection station is specified this inspection point will only be available at this inspection station.

Batch

Possibility to enter a corresponding batch.

Partial batch

Possibility to enter a corresponding partial batch.

Details

Generation of inspection points

If an inspection point is generated automatically on the basis of reaching a defined time or quantity interval, it has the flag "relating to time" or "relating to quantities". An inspection point that is assigned the flag "free" has been generated manually or on the basis of other events that make inspections become due.

Inspection points - editing functions

New inspection points cannot be created for completed or canceled inspection steps. The same applies to the modification of existing inspection points. To allow for an inspection point to be created or edited, the inspection step has to be released beforehand.

Inspection points - toolbar

Release

Function authorization: iripp.release

Transfers a completed inspection point to the "open" status.

Complete

Function authorization: iripp.complete

Opens a dialog window that allows for inspection points to be completed.

"Complete inspection point" detail application

Function authorization	iripp.complete

The corresponding button has to be clicked in the toolbar to complete an inspection point. Then the following editing dialog opens including an inspection result that cannot be changed and that is calculated on the basis of the inspection results for characteristics. In addition to this, in the tab “Evaluation” a usage decision for inspection points calculated on the basis of the inspection results for characteristics is preset, which may, however, be changed manually. To do so, there is a selection listof usage decisions for inspection points, filtered by the plant,, whereas every usage decision finally classifies the inspection point either as "pass" or "fail", irrespective of the inspection result. Additional information is displayed on the usage decision selected for the inspection point, once the selection has been confirmed. Additional information may, for example, be the code group, code number, catalog type and the factory/site.

MPDV services for customizing the system may provide further usage decisions for inspection points.

According to the configuration of identification fields, they can still be edited when inspection points are completed. If some identification fields require input (mandatory fields), they need to be filled out before inspection points can be completed. In addition to this, quantity fields may also be filled out.

"Inspection step/inspection point characteristics" detail application

Function authorization	iriscp.*

The detail application of inspection steps/inspection point characteristics is nearly identical to the master data of characteristics application. For this reason, reference is made only to additional features in the following.

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For further information on the definition of characteristics, please refer to the functional description of the document entitled MOC_CharacteristicsQM.

The inspection step characteristics are displayed below the inspection step within the master-detail grid. Provided that inspection points are in use and inspection points have been generated, the list of inspection points may be opened below the inspection steps. This list again allows for the characteristics pertaining to the selected inspection point to be opened.

The existing inspection plan determines which characteristics are included in the inspection step. In case the inspection plan is configured in a way specifying that a separate inspection step is to be generated for each operation (applying to several workplaces), the inspection step only includes the characteristics of a specific operation. Provided that the inspection plan has additionally been configured to generate a separate inspection step for every workstation, an inspection step always includes the characteristics of a fixed combination of operation and workplace.

An inspection point, in turn, only includes characteristics of the same event causing inspections to become due. Even the manual generation of inspection points represents a due date event. An inspection point is generated automatically, once e.g. a quantity interval defined for the inspection step characteristics has been achieved. This inspection point includes all characteristics that have the same quantity interval or that have a quantity interval that has the achieved quantity interval as least common multiple.

Example:
Characteristic 1 has a quantity interval of 100 and characteristic 2 of 200. Once the first 100 items have been produced, an inspection point is generated, which only includes characteristic 1. After another 100 items that have been produced, another inspection point is generated that includes the characteristics 1 and 2.

If an inspection becomes due because of changing the machine status, the generated inspection point only includes characteristics that have to be checked on the basis of the same machine status change.

Only manually generated inspection points are not limited with respect to the corresponding characteristics. These inspection points always include all characteristics of the related inspection step.

Inspection step/inspection point characteristics field descriptions

The paragraphs that follow only describe the fields that are available in addition to the characteristic master data or inspection plan characteristics.

Characteristic

Initial sample creation

Shows the field content transferred from the basic inspection plan characteristic. A separate sheet/page is printed for each category in the printed initial sample report.

Properties

Dynamically modified

This checkbox is enabled, provided that a characteristic is dynamically modified.

Dynamic modification

Determined inspection severity/inspection severity designation

This field shows the inspection severity determined for this characteristic on the basis of the dynamic modification rules specified within the inspection plan.

Revised inspection severity/inspection severity designation

This field only exists if dynamic modification is based on characteristics. A new initial inspection severity is displayed here, provided that the user has intervened in the process of dynamic modification and changed the current inspection severity (new initial inspection severity).

Please note

A note is displayed in this field, in case an error has occurred with the dynamic modification of this characteristic while inspection steps are generated.

Details

No cavity

The authorization "ipl_cav" is required for displaying these fields.

This option allows for a characteristic to be highlighted as irrelevant to cavities, although the respective inspection requirement specifies that the relevant characteristics are (actually) to be recorded in relation to cavities.

This checkbox has always to be enabled for attributive characteristics and inspection charts.

Sample group

This field specifies which sample group this characteristic belongs to.

Sampling

If a characteristic is assigned a sample group, this checkbox specifies whether or not a sample is to be generated by way of this characteristic. If this is the case, it is a "sampling characteristic". Consequently, the checkbox is to be enabled. A sample group is to be assigned to the sampling characteristic.

If these fields are not available, you require a new program version of this application.

Specifications

Sample size

This field shows the determined sample size, if within an inspection plan characteristic a sampling scheme is defined that does not require the sample size to be indicated, as it is calculated on the basis of the actual quantity of the inspection requirement. This is, for example, the case for the sampling schemes "batch inspection" or "100% inspection".

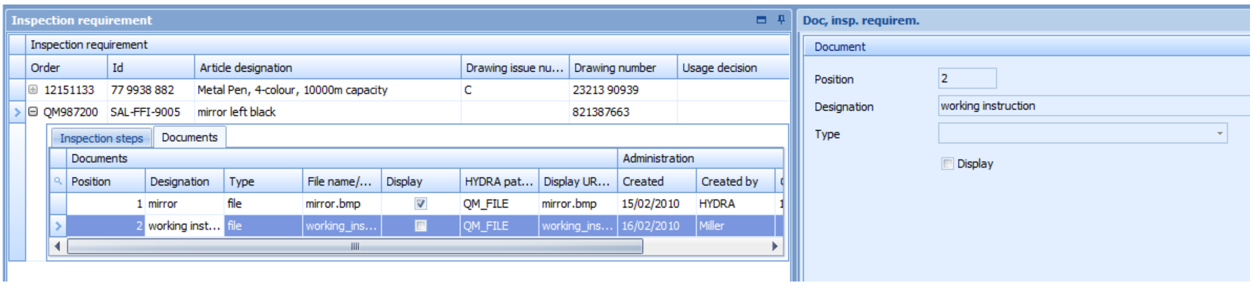
Go to

The other fields of the respective tabs correspond to those of the characteristic master data and are described in the documentation entitled "CAQ characteristic master data".

Inspection step/inspection point characteristics - editing functions

Inspection step or inspection point characteristics cannot be edited.

“Inspection request documents" and "characteristic documents" detail applications



The above screenshot shows how an inspection requirement document is assigned.

Provided that the "documents" tab has been activated, as many documents as required may be assigned to each inspection requirement and each inspection step characteristic within the master-detail grid. If this tab is activated in the master-detail grid corresponding editing buttons are enabled in the toolbar to edit the documents.

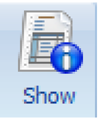
When documents are assigned, all formats registered by Windows are provided. Consequently, it is possible to assign simple documents (e.g. written in Word), drawings of any format and videos. However, the corresponding programs that are able to display the required formats have to be installed. In this context, the documents are opened by the program that has been linked in Windows.

The file types are "File", "URL" and "Text". The file name including the path may be entered manually for the "file" type. The "URL" file type enables access to the Internet or Intranet. The third file type "Text" allows for text to be entered directly.

A designation may be assigned to each defined document. Moreover, it may be determined in which order the documents are to be listed. The "position" field is used for this purpose (numeric input). The specifications made within this list must be unique. In addition, the checkbox "display with inspection" determines whether or not the document may be shown during the inspection process.

"Inspection plan documents“ and "documents for inspection plan characteristics" toolbar

In addition to the standard functions, there is also the button to show the documents.



Show documents

If a document link is defined this button opens and shows this document. However, a program, which can show the linked file type, has to be installed on the PC.

"Failure" detail application

A "failure list" detail application is provided on the level of inspection requirements. This list shows all failures of the type "failure type", "failure location", "failure cause" and "originator", which have been assigned to the characteristics, samples or measured values during the inspection process. The list even shows the failure types that have been generated automatically, e.g. "upper tolerance limit violated".

In addition to the respective failure, the following referenced data is also available.

- Inspection step
- Operation sequence number (OP sequence, AFO)
- Sample number
- Value number
- Failure type
- Characteristic number and designation
- Workplace number and designation
- Weighting (number, e.g. for an inspection chart characteristic)
- Comment,
- Failure date,
- Failure time

The list may be restricted in a flexible way by using the individual column filter. In addition, the grouping function allows for failures to be listed for every characteristic, for example.

Failure list - editing functions

Failure analysis entries can be created, edited or deleted for inspection requirements. If a new data record is created, further key information, such as

- Inspection step number,
- OP sequence number (uniquely identifies the characteristic),
- Sample number
- Value number

may be added to the assignment.

The target group can be found in the administrative area, as "administrative" skills are required for more precise assignment of this key information.

These authorizations are required.

Function authorization	failure.create => create new failure failure.edit => edit failure failure.delete => delete failure

Users must have profound knowledge of the inspection processes as they are not guided through the process of data collection/modification and validation checking is not performed.

The fields "area" and "inspection requirement number" can neither be modified nor assigned values during initial data creation. This data is taken from the previously selected inspection requirement when it comes to initial data creation.

The other fields can be assigned values as part of initial data creation. Only the fields "weighting" and "comment" may be changed in the editing mode.

"Measures" detail application

A "measure list" detail application is provided on the level of inspection requirements. This list shows all measures which have been assigned to the characteristics, samples or measured values during the inspection process.

In addition to the respective measure number and designation, the following data is also referenced.

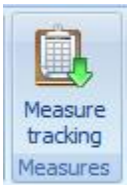
- Inspection requirement
- Inspection step
- Operation sequence number (OP sequence, AFO)
- Sample number
- Value number

- Measure type
- Party in charge incl. type
- Status
- Comment
- Text
- Effectiveness

The list may be restricted in a flexible manner by using the individual column filter. The grouping function provides further analysis options.

"Measures" toolbar

The button "measure tracking" can be used to open the measure tracking dialog to edit and create measures. Measure tracking is filtered by the internal and unique "serial number" of the measure in the "references" tab. Filtering is omitted if no measure is selected.



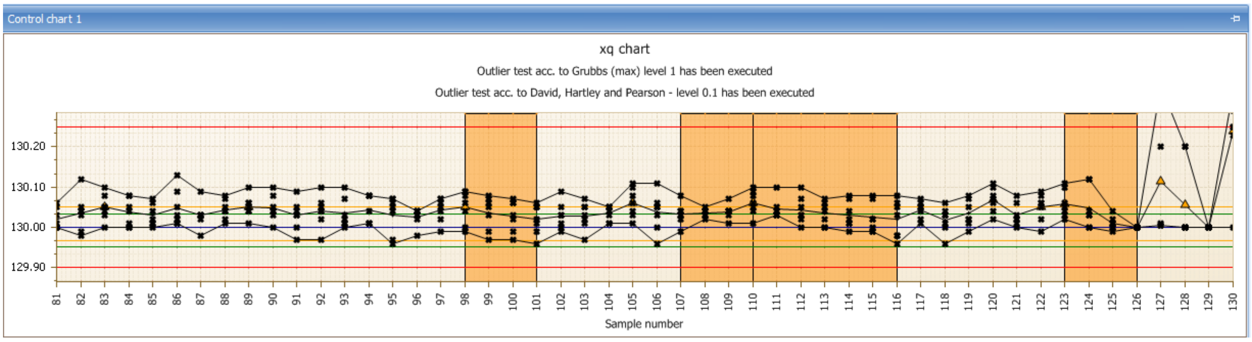
Go to measure tracking

Function authorization	failure.* => opens measure tracking

"Control chart 1 / 2" detail application

The control charts 1 and 2 defined for the inspection step characteristic are displayed by default. In case they are not defined for the inspection step characteristic, the xq and s charts are used for variable characteristics and the p and u chart for attributive characteristics.

All data of the selected inspection step characteristic is used as basis for graphic preparation.



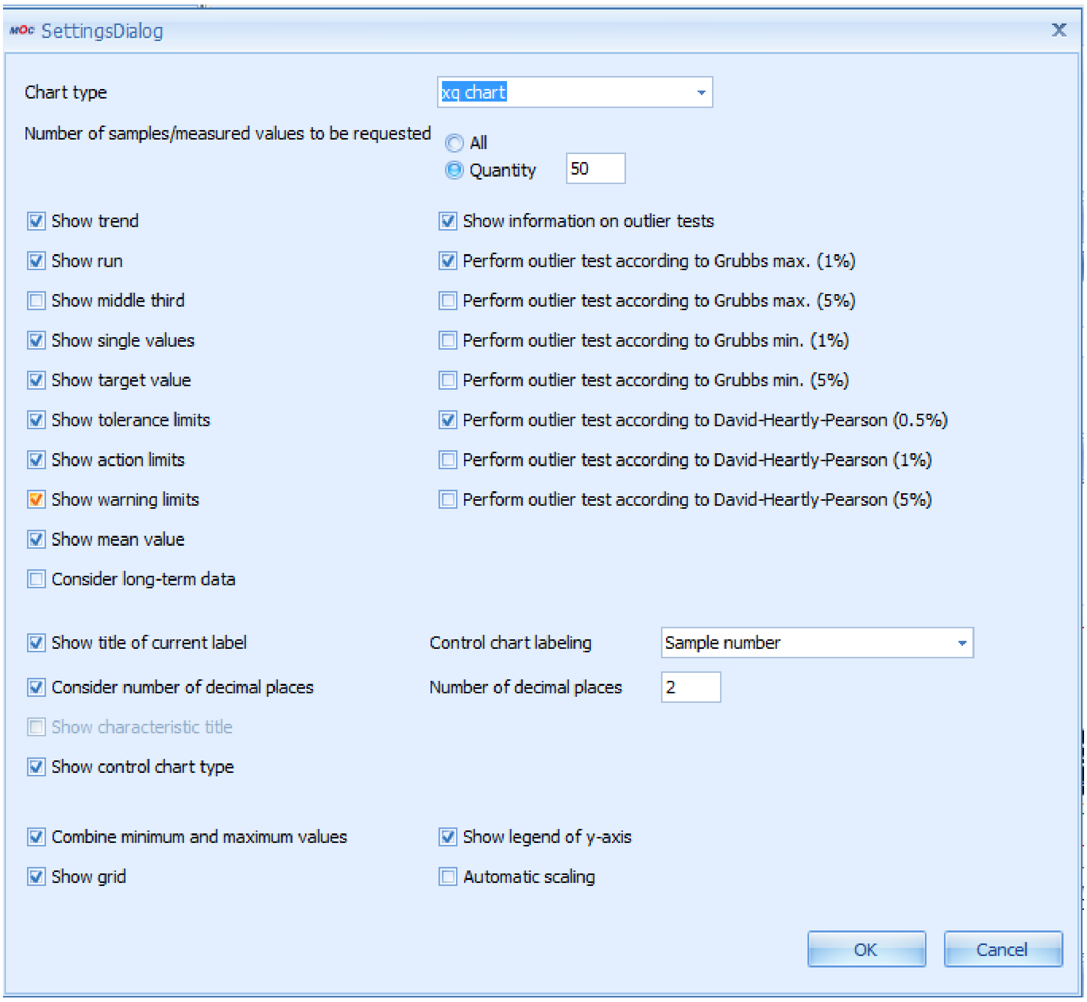
The contents of this application are defined by opening the dialog to configure "control chart 1" or "control chart 2". Changes made via this dialog are saved according to the user's requirements. Provided that no special settings have been made, the xq chart ("variable" characteristic type) or the p chart ("inspection chart" or "attributive" characteristic type) is displayed, by default. Subject to the characteristic type, the user may choose to display one of the following control charts by default.

Variable characteristic

- Xq chart
- s chart
- R chart
- Single value chart
- Median chart

Attributive characteristic

- p chart
- np chart
- c chart
- u chart



The paragraphs that follow explain the essential configuration options.

Number of the samples to be requested

Specifies how many samples or single measured values are to be displayed in the control chart.

Display …

The options that include the term "display ..." enable or disable the presentation of the corresponding data in the control chart.

Consider long-term data

This option has to be checked to integrate archived data from the medium-term data area.

Combine minimum and maximum values

It is recommendable to show the minimum and maximum values that are each connected by a line to improve the presentation of the range of dispersion of single values.

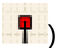
Automatic scaling

The "automatic scaling" function allows for all values to be displayed, irrespective of the existing limit values. This shows even extreme outliers in the control chart. The disadvantage is that there is less space for the other measured values and, it is very often the case that a changing value pattern can hardly be recognized.

Show trend / run / middle third

The monitoring functions "trend", "run" and "MiddleThird" allow better surveillance of a process than by using the control chart alone. However, they can only be used with control charts for xq and median. The trend allows visualization of an upward or downward tendency in the process over several samples. By default, these are seven subsequent rising or falling values. The run shows sections in which the process runs above or below the mean value (when displayed, otherwise the target value) over several samples. By default a run is recognized if seven subsequent values are above the mean value. The number of seven samples/values for recognizing a trend or run, which is set by default, may be changed by customizing the system. "MiddleThird" refers to an unusually high or low number of values, in the section of the control chart viewed, within the middle third of the area bounded by the action limits.

Respective analyses are performed automatically for "trend", "run" and "MiddleThird". The control chart shows in a graphic if a trend, run and / or MiddleThird is available/detected. The following events are altogether represented by icons or color codes in the control chart.

- Trend (can be recognized by a colored area)
- Run (can be recognized by a colored area))
- Middle Third (can be recognized by a colored area)
- Outlier (icon: )
- Xq violates action limit (icon )

The presentation of outliers is restricted to the xq chart and median chart and connected with the presentation of single values. There are different functions for performing and presenting outlier tests. The different outlier tests for the different levels may be activated separately. Provided that the presentation of outlier tests has been activated, the result of this test is displayed in text form above the corresponding control chart.

The below outlier tests are available for the different inspection levels, whereas the inspection level is indicated in parentheses.

- Grubbs max. (1 %)
- Grubbs max. (5 %)
- Grubbs min. (1 %)
- Grubbs min. (5 %)
- David-Hartley-Pearson (0.5 %)
- David-Hartley-Pearson (1 %)
- David-Hartley-Pearson (5 %)

Notes on outlier tests

- The outlier tests do not refer to the collectivity of all samples, i.e. every sample is considered individually. Consequently, the following phenomena may appear:
  - Despite a large range between minimum and maximum value no outlier can be identified. A reason for this may be the equal distribution of the single values within the sample.
  - Despite a low range between minimum and maximum value an outlier can be identified. A reason for this may be an accumulation of many values at one “point” so that an individual value having a certain distance to this agglomeration is identified as outlier within the sample.
- At least three values have to be in the sample in order to be able to perform an outlier test. The more values are available within a sample the more uniform the general view of the outliers becomes compared to all samples.
- The Grubbs outlier test is performed with a sample size of 2 < n < 148 (n = number of values within a sample). The outlier test according to David, Hartley und Pearson is performed at a sample size of 2 < n < 1251 (n = number of values within a sample).

X-axis labeling

The below information is available to label the x-axis of control charts.

- Sample number
- Order number
- PPS reference number
- Purchase order number
- Batch
- Article number
- Machine number
- Cavity number
- Date + time of the first measured value of a sample
- Date + time of the last measured value of the sample
- Date + time of sample completion
- Badge number of the first measured value of the sample
- Badge number of the last measured value of the sample
- Badge number of sample completion

If these fields are not available, you require a new program version of this application.

- Partial batch
- Workplace
- Production workplace

- Field 1
- Field 2
- Field 3
- Field 4
- Field 5
- Field 6
- Field 7
- Field 8

The fields "field 1" to "field 8" are enabled subject to their usage by MPDV customizing and assigned an individual designation. As these field names are flexible, they are only entitled "field 1" to "field 8" in this document. Field 1 includes, for example, the tool if cavity-related data collection is enabled.

Tool tips within the control chart

When a value is labeled being an outlier (red rhomb) detailed information on which test(s) was (were) the crucial factor for this determination is displayed when going with the mouse over this labeling (rhomb).

For mean values the tool tip shows the value, date and time for the first and last measured value of this sample as well as the corresponding inspection request number and inspection step number.

The exact value is shown for single measured values.

A special note indicating why an area is colored is also displayed for the colored areas when a trend, run or middlethird is recognized.

Sorting of measured values

The type of control chart sorting is determined by the server and depends on how the control chart filters are configured. If the filters "operation sequence" and "inspection request number" or "operation sequence" and "inspection step number" are indicated the characteristic is identified uniquely and sorting is based on the sample number.

Sorting is based on date and time if either the filter field "operation sequence" is left empty or it is filled out and the fields "inspection request number" and "inspection step" number are left empty.

The optical presentation of trend or run always refers to how the measured values are sorted. However, the automatic generation of the failure type "trend" is always based on the measured values being sorted by the sample number, as the numbers for the operation sequence, inspection request and the inspection step are always known at the time data is recorded.

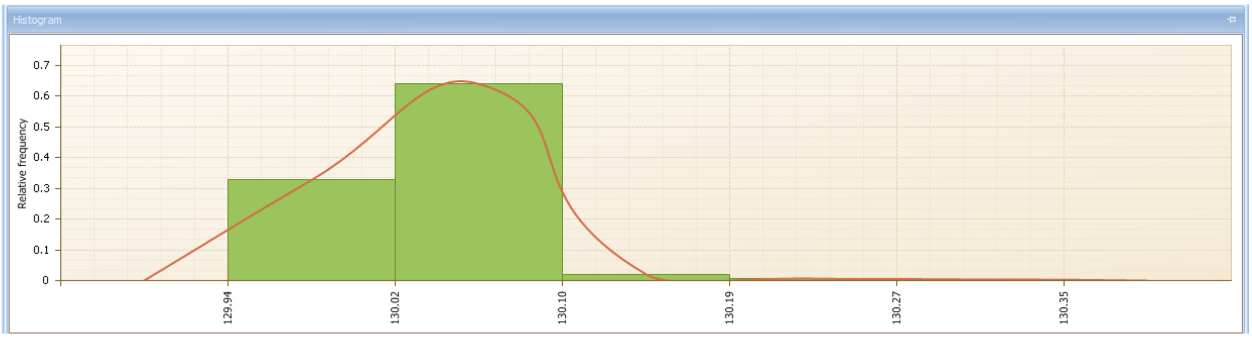
Consequently, the following scenario might be possible.

- The control chart is sorted by date and time and a trend is shown, although the automatic failure "trend" has not been generated.

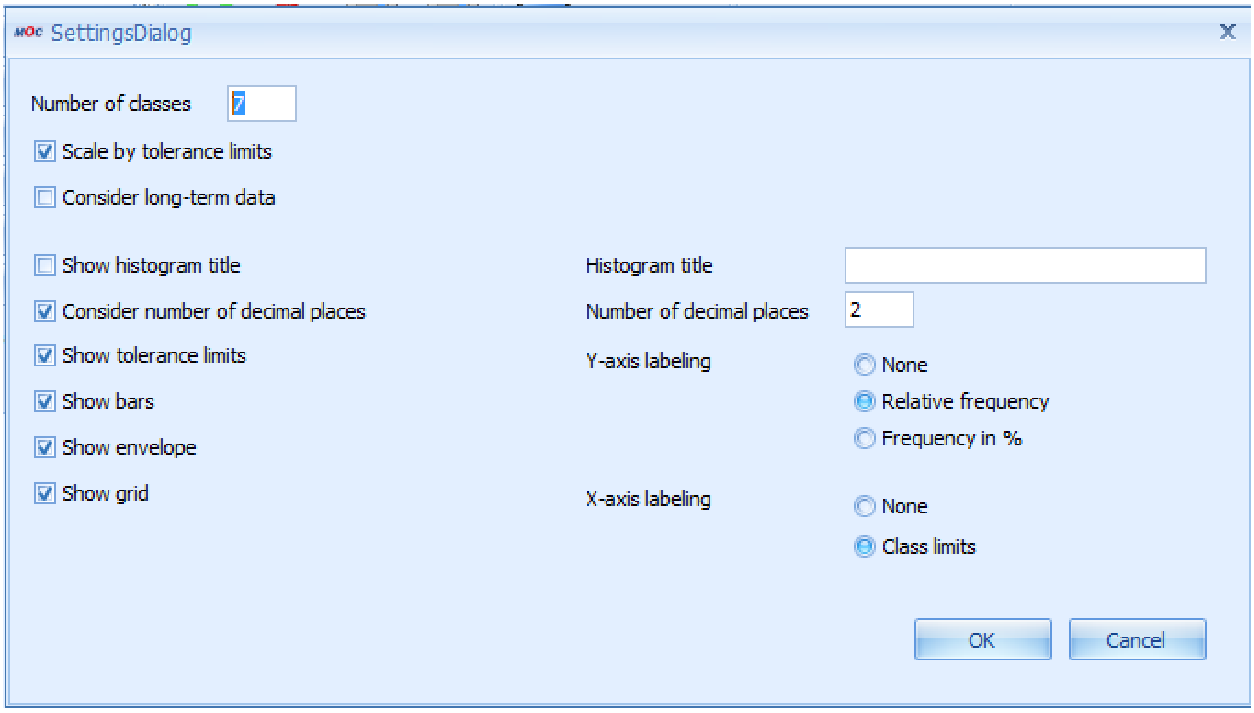
The control chart is sorted by the sample number and no trend is shown, although the automatic failure "trend" has been generated.

"Histogram" detail application

All data of the selected inspection step characteristic is used as basis for graphic preparation.



Unlike the control charts, which display a specified excerpt from the sample set, the histogram is always based on the entire set of available samples matching the selection filter criteria. The appearance of the histogram is determined by the number of classes and by any elements additionally displayed. The contents of this application are defined by opening the dialog to configure the "histogram". Changes made via this dialog are saved according to the user's requirements.



The paragraphs that follow explain the essential configuration options.

Number of classes

Determines the number of histogram classes according to which the measured values are to be distributed. If represented within the tolerance limits (option "scale by tolerance limits") one histogram class each is outside of the tolerance limits.

Scale by tolerance limits

Enabled: The classes are in between the tolerance limits with each one "outlier class" to the left and to the right.

Disabled: The classes include the range of all measured values (no separate classes for values outside of the tolerance limits).

Consider long-term data

Includes the archived data from the medium-term data area.

Show histogram title

If a special title is to be displayed it may be entered by enabling this option.

x-axis labeling

The x-axis shows the corresponding values of the class limits if the "class limits" option is set. The number of decimal places that are to be displayed may be set by the two configuration options for decimal places.

Consider the number of decimal places

Takes into account the defined number of decimal places in the x-axis labeling.

Control chart 1/2 and histogram

Control chart 1 **Control chart 1 settings**

Opens a dialog to configure the settings of control chart 1. The corresponding details are described in the respective detail application.

Control chart 2 **Control chart 2 settings**

Opens a dialog to configure the settings of control chart 2. The corresponding details are described in the respective detail application.

Histogram **Histogram settings**

Opens a dialog to configure the histogram settings. The corresponding details are described in the respective detail application.

"Samples" detail application

The statistical key figures about every single sample can be found in the "samples" detail application on the level of inspection step characteristics. In addition to the referenced key fields, such as the inspection requirement number, inspection step number and sample number, provided that they are available or can be calculated, the key figures

- Xq
- Xq floating
- R floating
- Standard deviation s
- s floating
- Minimum
- Maximum
- Range
- Median
- P
- U
- Number of defects

are listed. The list additionally shows the characteristic specifications (tolerance, action and warning limits) as well as the referenced machine.

"Samples" toolbar

Complete sample

It might be necessary to complete samples manually, as measured values can be entered on "administration" level or as there might be inspections without inspection points including the creation of new samples and/or inspections without having reached the sample size. Samples can even be completed if they have not been completed by AIP inspection data collection.

Release sample

This function enables samples to be released once more and the collection of additional measured values for this sample number, provided the defined sample size has not yet been reached.

"Single values" detail application

The single measured values for all variable inspection step characteristics can be found in the "single values" detail application on the level of inspection step characteristics. For attributive characteristics the following information

- Number of defects
- Number of NCU (non-conforming units) and
- Defects

is displayed. Moreover, it can be recognized whether the measured value or the attributive assessment is valid or invalid.

The characteristic specifications (tolerance, action and warning limits) are listed in addition to the referenced key fields, such as the inspection requirement number, inspection step number, sample number, and value number. Every entry also shows the date and time when it was recorded and edited as well as the responsible user.

Single values - editing functions

Measured values can be created, edited or deleted for a variable inspection step characteristic and/or inspection point characteristic and the number of non-conforming units may be created, edited or deleted for attributive characteristics on "administration" level. If a new data record is created, further key information, such as

- the sample number
- and value number

may be added to specify the assignment.

The target group can be found in the administrative area, as "administrative" skills are required to provide for more precise assignment of this key information. Users must have profound knowledge of the inspection processes, as they are not guided through the process of data collection/modification and validation checking is not performed.

These authorizations are required.

Function authorization	value.create => create new measured value value.edit => edit measured value value.delete => delete measured value

Users must have profound knowledge of inspection processes, as they are not guided through the process of data collection/modification and validation checking is not performed.

The fields

- Area,
- Inspection request number,
- Inspection step no.,
- Inspection step number (only available for inspection point characteristics),
- OP sequence (uniquely identifies the characteristic),
- Upper tolerance limit,
- Target value,
- Lower tolerance limit and
- Number of decimal places

can neither be modified nor assigned values during initial data creation. When it comes to initial data creation, this data is taken from the previously selected characteristic and the referenced inspection requirement as well as from the inspection point.

The other fields can be assigned values and/or changed during initial data creation.

The field "single value of cavity" is only visible if

- the inspection requirement shows an ID indicating that data is generally collected in relation to cavities and
- the relevant characteristic is not assigned an ID indicating that it still has to be checked without relation to cavities.

If a new data record is created, the field for the measured value will be restricted to the number of decimal places. The other decimal places are filled up with zeros. For technical reasons, these "additional" decimal places are shown during processing.

The field "measured value" is shown for variable characteristics. Instead of the measured value field, the fields "sample size (checked)", and "number of DU" are shown for attributive characteristics, i.e. this also includes inspection chart characteristics.

"Statistics" detail application

The common statistical key figures about the selected characteristic can be found in the "statistic" detail application on the level of inspection step characteristics.

For variable characteristics these are

- Number of samples
- Number of measured values
- Xqq
- Minimum
- Maximum
- R
- S
- S relative
- Sigma
- Cp and
- Cpk

For attributive characteristics these are

- Number of non-conforming units

- Number of defects
- p and
- u.